

Experiment - 4.1.1. Set Operations

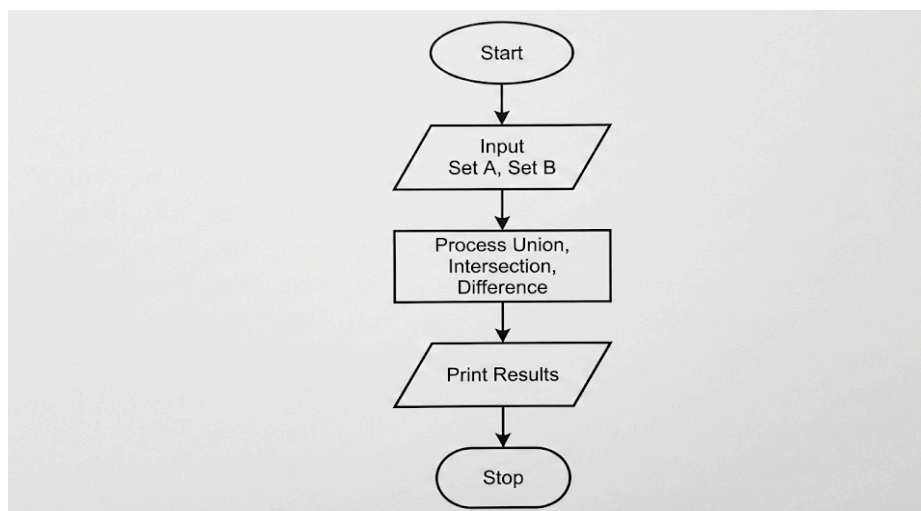
1. Aim

To design and implement a Python program that performs mathematical set operations, specifically union, intersection, and difference, on two sets: Set A and Set B. The program prompts the user to input space-separated integers for both sets and computes the results.

2. Pseudocode

1. **START**
2. **PRINT** the prompt "Set A: " and **READ** a line of space-separated integers.
3. **CONVERT** the input values into a mathematical set and **STORE** it in the variable A.
4. **PRINT** the prompt "Set B: " and **READ** a line of space-separated integers.
5. **CONVERT** the input values into a mathematical set and **STORE** it in the variable B.
6. **CALCULATE** the union of Set A and Set B ($A \cup B$) and **STORE** the result in union_set.
7. **CALCULATE** the intersection of Set A and Set B ($A \cap B$) and **STORE** the result in intersection_set.
8. **CALCULATE** the difference between Set A and Set B ($A - B$) and **STORE** the result in difference_set.
9. **PRINT** the text "Union:" followed by the value of union_set.
10. **PRINT** the text "Intersection:" followed by the value of intersection_set.
11. **PRINT** the text "Difference:" followed by the value of difference_set.
12. **END**

3. Flowchart



4. Python Program

Read sets

```
A = set(map(int, input("Set A: ").split()))
```

```
B = set(map(int, input("Set B: ").split()))
```

Operations

```
union_set = A | B
```

```
intersection_set = A & B
```

```
difference_set = A - B
```

Output

```
print("Union:", union_set)
```

```
print("Intersection:", intersection_set)
```

```
print("Difference:", difference_set)
```

5. Experiment Screenshot

The screenshot displays the C++DE TANTRA IDE interface. On the left, the problem description for "4.1.1. Set Operations" is visible, detailing the input format (two sets A and B), output format (Union, Intersection, and Difference), and sample test cases. The main editor shows the Python code implementing these operations. The right sidebar displays the execution results, including a table of performance metrics and a list of test cases.

Problem Description:

Write a Python program to perform union, intersection and difference operations on *Set A* and *Set B*.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for *Set A*
- The second input prompts "Set B: " followed by space-separated list of integers for *Set B*

Output Format:

- The first line prints "Union: " followed by the union of *Set A* and *Set B*
- The second line prints "Intersection: " followed by the intersection of *Set A* and *Set B*
- The third line prints "Difference: " followed by the difference of *Set A* and *Set B*

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases:

Test case 1

Set A: 0 2 4 5 8
Set B: 1 2 3 4 5
Union: {0, 1, 2, 3, 4, 5, 8}
Intersection: {2, 4, 5}
Difference: {0, 8}

Test case 2

Set A: 10 11 22 33 44 55
Set B: 15 16 17 18 19 24
Union: {10, 11, 15, 16, 17, 18, 19, 22, 24, 33, 44, 55}
Intersection: set()
Difference: {10, 11, 22, 33, 44, 55}

Execution Results:

Metric	Value
Average time	0.008 s
Maximum time	0.009 s
Test Case Status	2 out of 2 shown test case(s) passed 2 out of 2 hidden test case(s) passed

Test Case 1 Details:

Expected output:
Set A: 0 2 4 5 8
Set B: 1 2 3 4 5
Union: {0, 1, 2, 3, 4, 5, 8}
Intersection: {2, 4, 5}
Difference: {0, 8}

Actual output: